

1. $I = |q| n A \bar{v}$ where I is the conventional current

Now $|q| = e$ and since $\rho = \frac{m}{V}$ and $n = \frac{N}{V}$, we have that $n = \frac{\rho N}{m}$ (where N is the number of elec)

$$\text{Now } m = N \cdot m_e = N \cdot \frac{M}{N_A}$$

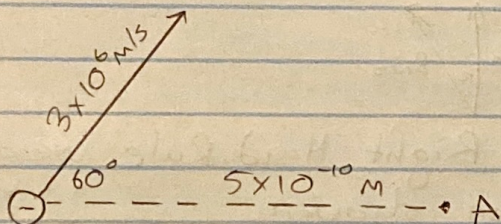
$$\text{Thus } n = \frac{\rho N}{N \cdot \frac{M}{N_A}} = \frac{\rho N_A}{M}$$

$$\Rightarrow n = \frac{8.92 \times 6.02 \times 10^{23}}{63.54} = 8.45 \times 10^{22}$$

$$I = e(8.45 \times 10^{22})(1 \times 10^{-3})^2 \bar{v}$$

$$\bar{v} = 74.0 \text{ m} \cdot \text{s}^{-1}$$

2a)



$$\hat{E} = \langle -1, 0, 0 \rangle$$

$\hat{B} = \langle 0, 0, +1 \rangle$ by the right hand rule ($q < 0$, so $\hat{B}$ flips)

$$b) E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} = 9 \times 10^9 \cdot \frac{1.6 \times 10^{-19}}{(5 \times 10^{-10})^2} = 5.76 \times 10^9 \text{ N/C}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{q \vec{v} \times \hat{r}}{r^2}$$

$$\therefore B = \frac{\mu_0}{4\pi} \cdot \frac{q}{r^2} v \sin \theta = 10^{-7} \cdot \frac{1.6 \times 10^{-19}}{(5 \times 10^{-10})^2} (3 \times 10^6) \sin 60 = 0.166 \text{ T}$$

$$3a) \vec{B}_{\text{total}} = \vec{B}_{\text{right}} + \vec{B}_{\text{left}}$$

Between the loops, $\hat{B}_{\text{right}} = \langle 1, 0, 0 \rangle$ and $\hat{B}_{\text{left}} = \langle 1, 0, 0 \rangle$

Also $B_{\text{right}} = B_{\text{left}}$, so $\vec{B}_{\text{total}} = 2 B_{\text{right}} \hat{i}$

$$\text{Now } B_{\text{right}} = \frac{\mu_0}{4\pi} \cdot \frac{2 I \pi R^2}{(z^2 + R^2)^{3/2}} = 10^{-7} \times \frac{2 \times 2 \times (\pi \times 0.03^2 \times 10)}{(0.1^2 + 0.03^2)^{3/2}} = 4.94 \times 10^{-7} \text{ T}$$

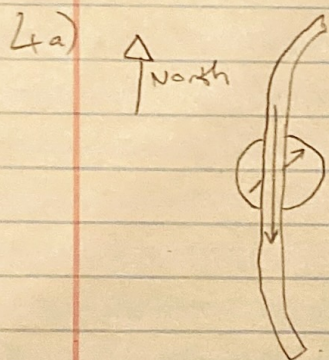
$$\text{Thus } \vec{B}_{\text{total}} = \langle 1.99 \times 10^{-5}, 0, 0 \rangle \text{ T}$$

b) Using the approximation, $B_{right} = \frac{\mu_0}{4\pi} \cdot \frac{2I\pi R^2}{z^3}$
 $= 10^{-7} \cdot \frac{2 \times 2 \times (\pi \times 0.03^2 \times 10)}{0.13^3}$
 $= 1.13 \times 10^{-5} T$

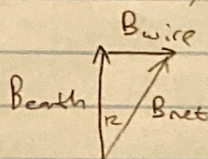
So $\vec{B}_{net} = \langle 2.26 \times 10^{-5}, 0, 0 \rangle T$

The percentage error is 13.57%. This is a bad approximation

c) In this case, $\hat{B}_{right} = \langle -1, 0, 0 \rangle$, so $\vec{B}_{total} = \vec{0}$

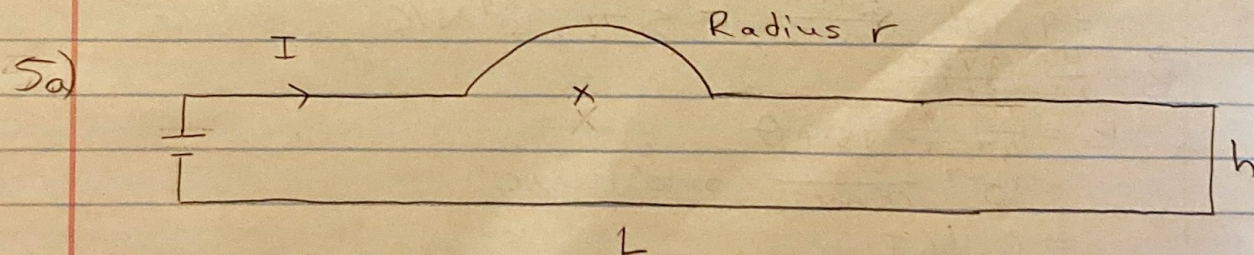


For $\vec{B}_{net}$, we need:



Using Right Hand Rule, we get direction shown.

b) We have that $\tan 12 = \frac{B_{wire}}{B_{earth}} = \frac{B_{wire}}{2 \times 10^{-5}} \Rightarrow B_{wire} = 4.25 \times 10^{-6}$
 Thus $\frac{\mu_0}{4\pi} \cdot \frac{2I}{r} = 4.25 \times 10^{-6}$ since $r \ll L$
 $\Leftrightarrow 4.25 \times 10^{-6} = 10^{-7} \cdot \frac{2I}{3.6 \times 10^{-3}}$
 $\Leftrightarrow I = 0.0765 A$



$\hat{B} = \langle 0, 0, -1 \rangle$ (into the page)

b) All around the wire, $\hat{B} = \langle 0, 0, -1 \rangle$
 Now assuming $\vec{B}_{net} = \vec{B}_{semi} + \vec{B}_{bot wire}$ and thus
 $B_{net} = B_{semi} + B_{bot wire}$

Now $B_{\text{semi}} = \frac{1}{2} B_{\text{ring}}$ by symmetry

$$\begin{aligned} \text{so } B_{\text{semi}} &= \frac{1}{2} \cdot \frac{\mu_0}{4\pi} \cdot \frac{2I\pi r^2}{(r^2 + r^2)^{3/2}} = \frac{1}{2} \cdot \frac{\mu_0}{4\pi} \cdot \frac{2I\pi r^2}{2\sqrt{2}r^3} \\ &= \frac{1}{2} \cdot \frac{\mu_0}{4\pi} \cdot \frac{I\pi}{\sqrt{2}r} \\ &= \frac{\mu_0}{4\pi} \cdot \frac{I\pi}{2\sqrt{2}r} \end{aligned}$$

$$\text{Also } B_{\text{wire}} = \frac{\mu_0}{4\pi} \frac{LI}{\ln(h^2 + (\frac{L}{2})^2)}$$

$$\text{Thus } B_{\text{total}} = \frac{\mu_0}{4\pi} \left[\frac{LI}{\ln(h^2 + (\frac{L}{2})^2)} + \frac{I\pi}{2\sqrt{2}r} \right]$$

$$6a) \vec{E}_{\text{proton}} = \frac{1}{4\pi\epsilon_0} \cdot \frac{e}{r^2} \hat{i}$$

$$\vec{B}_{\text{proton}} = \frac{\mu_0}{4\pi} \cdot \frac{qv \sin\theta}{r^2} (-\hat{k})$$

$$= \frac{\mu_0}{4\pi} \cdot \frac{e v_f \sin\theta}{r^2} (-\hat{k})$$

$$\begin{aligned} \text{Clearly then } F_x &= qE_{\text{proton}} \\ &= \frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{r^2} \text{ in direction } (-\hat{i}) \end{aligned}$$

$$\text{and } F_y = 0$$

$$b) \text{ Again, } \vec{E}_{\text{elec}} = \frac{1}{4\pi\epsilon_0} \cdot \frac{e}{r^2} (\hat{i}) \text{ and}$$

$$\vec{B}_{\text{elec}} = \frac{\mu_0}{4\pi} \cdot \frac{e v_e}{r^2} \hat{k}$$

$$\text{So } F_x = \frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{r^2} \text{ in direction } \hat{i} \text{ and } F_y = 0$$

$$\begin{aligned} c) \text{ Yes, } \vec{F}_{\text{on elec}} &= \left\langle -\frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{r^2}, 0, F_z \right\rangle \text{ and } \vec{F}_{\text{on prot}} \\ &= \left\langle \frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{r^2}, 0, F_z \right\rangle \text{ so this does hold for} \\ &F_x \text{ and } F_y \end{aligned}$$

d) Yes, for the system of both the particles, there is no external force (all forces are internal) so momentum is conserved